Inorganic Cumulative Exam

October 11, 2013

At one time perchlorate was used extensively as a non-coordinating ligand in studies of transition metal complexes. This practice was abandoned. The collective memory of the chemical community seems to be about 20 years and some of the critically important implications are being forgotten.

1. Give the formula for perchlorate ion.

2. Why was this practice abandoned?

3. Some of the initial studies of the coordinating ability of perchlorate were done with simple nickel complexes. IR spectroscopy was used heavily in these studies. Describe in detail how IR spectroscopy can be used to determine the bonding or the lack of bonding of perchlorate ion to Ni^{2+}. Your discussion should include how group theory is applied to this problem.
   a. What type of sample preparation/technique would be appropriate for the IR investigation and why?
   b. Should water be excluded in these investigations?

4. Beside IR and Raman spectroscopy what other tool might be useful in establishing the existence of perchlorate coordination. Explain how and include what specifics you would look for. You can assume that suitable crystals are not available for diffraction studies. If you cannot think of one discuss why a particular spectroscopic technique will not work.